

***Antiproliferative effects of fresh and thermal
processed green and red cultivars of curly kale
(Brassica oleracea L. convar. acephala var.
sabellica).***

Brassica vegetables contain a diverse range of phytochemicals with biological properties such as antioxidant and anticancer activity. However, knowledge about how biological activities are affected by processing is lacking. A green cultivar and a red cultivar of curly kale were evaluated for water/methanol-soluble phytochemicals before and after processing involving blanching, freeze storage, and boil-in-bag heat treatment. In both kale cultivars, processing resulted in a significant decrease of total phenolics, antioxidant capacity, and content and distribution of flavonols, anthocyanins, hydroxycinnamic acids, glucosinolates, and vitamin C. Interestingly, the red curly kale cultivar had a higher capacity to withstand thermal loss of phytochemicals. The extracts of both green and red curly kale inhibited the cell proliferation of three human colon cancer cell lines (Caco-2, HT-29, and HCT 116). However, extracts from fresh plant material had a significantly stronger antiproliferative effect than extracts from processed plant material.